FANFEI LI

fanfei.fli@gmail.com · Website · Google Scholar · LinkedIn

Second year Machine Learning PhD student interested in knowledge acquisition, data-efficient learning, and generalization in AI models, with a focus on data-centric approaches to pretraining and self-improvement.

EDUCATION

Max Planck Institute for Intelligent Systems Jun 2025 – Expected 2028
PhD in Machine Learning, University of Tübingen
Advisor: Prof. Wieland Brendel

University of Tübingen Sep 2022 – May 2025
M.S. in Quantitative Data Science Methods (Grade: 1.2, German scale; 1.0 best)

UCLA Anderson School of Management Sep 2019 – Dec 2020
M.S. in Business Analytics (GPA: 3.8/4.0)

University of California, Los Angeles Sep 2015 – Jun 2019
B.S. in Applied Mathematics, Minor in Music Industry
College Honors (UCLA's highest undergraduate academic distinction); Dean's Honors List

SELECTED PUBLICATIONS

[1] **LittleLearner: Language Models Under Pedagogically Controlled Knowledge Exposure** ***Oral: NeurIPS 2026***
F Li, J Zeller*, M Prada-Corral, T Wiedemer, P Mayilvahanan, R Cotterell, W Brendel*

[2] **MentisOculi: Revealing the Limits of Reasoning with Mental Imagery** *ICML 2026*
J Zeller, T Wiedemer, F Li, T Klein, P Mayilvahanan, M Bethge, F Wichmann, R Cotterell, W Brendel

[3] **LAION-C: An Out-of-Distribution Benchmark for Web-Scale Vision Models** *ICML 2025*
F Li, T Klein, W Brendel, R Geirhos, R S Zimmermann

* Equal contribution

EXPERIENCE

Max Planck Institute for Intelligent Systems Oct 2023 – May 2025
Research Intern Tübingen, Germany

- Developed LAION-C, an out-of-distribution robustness benchmark with six novel synthetic distortions designed to challenge vision models trained on web-scale data.
- Evaluated 50+ vision models across different architectures and training regimes, comparing their performance with human observers to investigate discrepancies in out-of-distribution robustness.

TikTok Nov 2020 – Oct 2022
Data Scientist (full-time from March 2021) Beijing, China

- Applied time-series clustering to analyze creator growth trajectories, identify behavioral patterns, and inform data-driven product strategies.
- Used causal inference methods (PSM and difference-in-differences) to quantify the impact of individual and organizational factors on creator growth and monetization.
- Built and maintained large-scale data pipelines in SQL and Python, providing reliable datasets and metrics for downstream analysis and modeling.

UCLA Anderson School of Management & Riot Games Jul 2020 – Aug 2020
Research Project Los Angeles, CA

- Ran experiments with XGBoost and LSTM models on minute-level League of Legends gameplay data to estimate in-game win probabilities.
- Used Python to preprocess 300GB of data, performed feature selection, feature engineering, and hyperparameter tuning.

Amazon Web Services May 2020 – Sep 2020
Data Engineer Intern Beijing, China

- Built and deployed production ETL pipelines to ingest data from Amazon S3 into Redshift, including modifications to Java and Ruby code to migrate storage endpoints between AWS regions.
- Developed automated data processing and reporting workflows in SQL and Python to support analysis of subscription, revenue, and customer behavior.